

VPN and IPsec

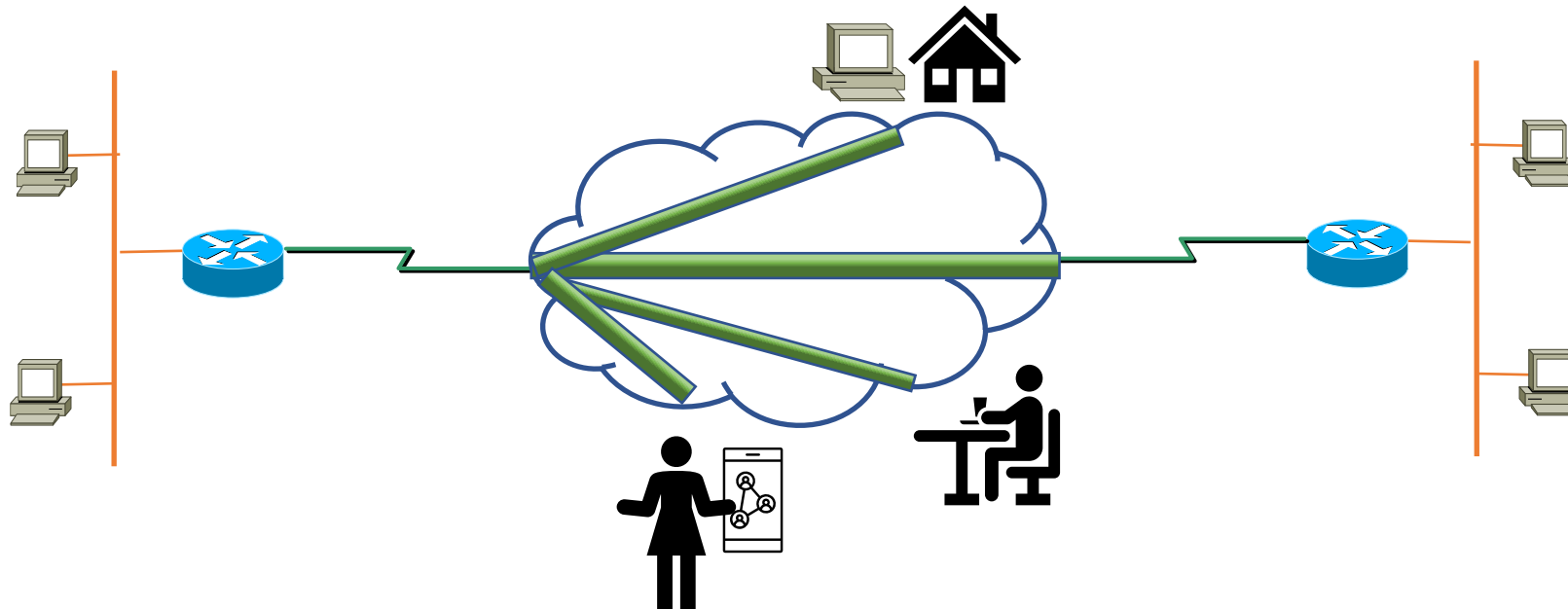
VPN and IPsec Concepts

VPN Technology

VPN Technology

What is a VPN?

A **Virtual Private Network (VPN)** is a set of private connections transported over a public network.



What is a VPN?

Cont'd

VPN concept has changed over time. Current VPN definition can be stated as:

A virtual private network, or VPN, is an encrypted connection over the Internet from a device to a network (or between sites). The encrypted connection helps ensure that sensitive data is safely transmitted. It prevents unauthorized people from eavesdropping on the traffic and allows the user to conduct work remotely. VPN technology is widely used in corporate environments.

Benefits of a VPN

Major Benefits of a VPN are shown in the table.

Benefit	Description
Cost Savings	Organizations can use VPNs to reduce their connectivity costs while simultaneously increasing remote connection bandwidth.
Security	Encryption and authentication protocols protect data from unauthorized access.
Scalability	VPNs allow organizations to use the internet, making it easy to add new users without adding significant infrastructure.
Compatibility	VPNs can be implemented across a wide variety of WAN link options including broadband technologies. Remote workers can use these high-speed connections to gain secure access to corporate networks.

How a VPN works

The general process of a VPN involves the use of “VPN Gateways” and / or “VPN Clients”

It includes tunneling and data encoding as it travels between a Client and a Gateway, which then decrypts the data and sends it on to its destination.

Internet Protocol Security (IPSec) and Secure Sockets Layer (SSL) are tunnel and authentication Technologies used in VPN deployments.

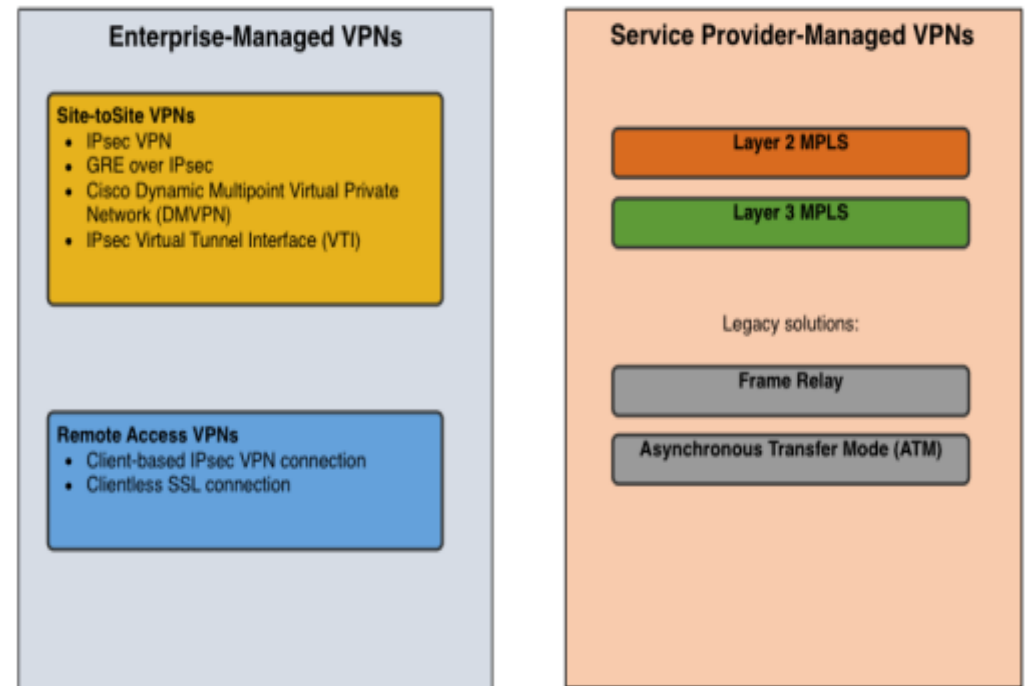
- Internet Protocol Security (IPsec) connections use pre-shared keys on clients and servers in order to encrypt and send traffic back and forth.
 - Secure Sockets Layer (SSL) use public keys. SSL VPN function is built-in web browsers.
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VPN Technology

Enterprise and Service Provider VPNs

VPNs can be managed and deployed as:

- **Enterprise VPNs.** Common solution for securing enterprise traffic across the internet. Site-to-site and remote access VPNs are created and managed by the enterprise using IPsec and SSL VPNs
- **Service Provider VPNs.** Created and managed by the provider network. The provider uses Multiprotocol Label Switching (MPLS) at Layer 2 or Layer 3 to create secure channels between an enterprise's sites, effectively segregating the traffic from other customer traffic.



VPN Types

VPN Types

VPN Types

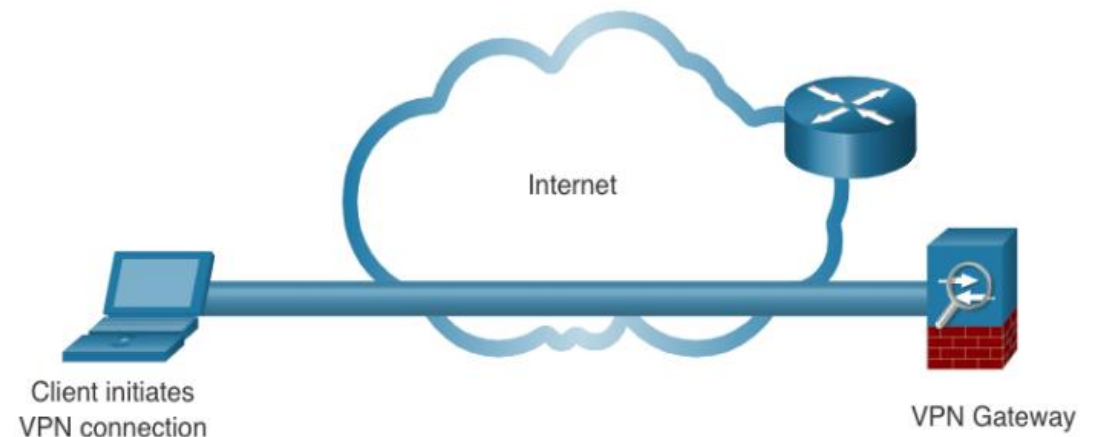
Remote-Access VPN

Remote-Access VPN

A Remote-Access VPN securely connects devices outside the corporate office. These devices are known as endpoints and may be laptops, tablets, or smartphones.

Remote-access VPNs are typically enabled dynamically by the user when required and can be created using either IPsec or SSL.

Increasingly, advances in VPN technology enable security checks to be carried out to ensure the device is secure before it is granted permission to connect.



Remote-Access VPN

Cont'd

Clientless VPN connection. The connection is secured using a web browser SSL connection.

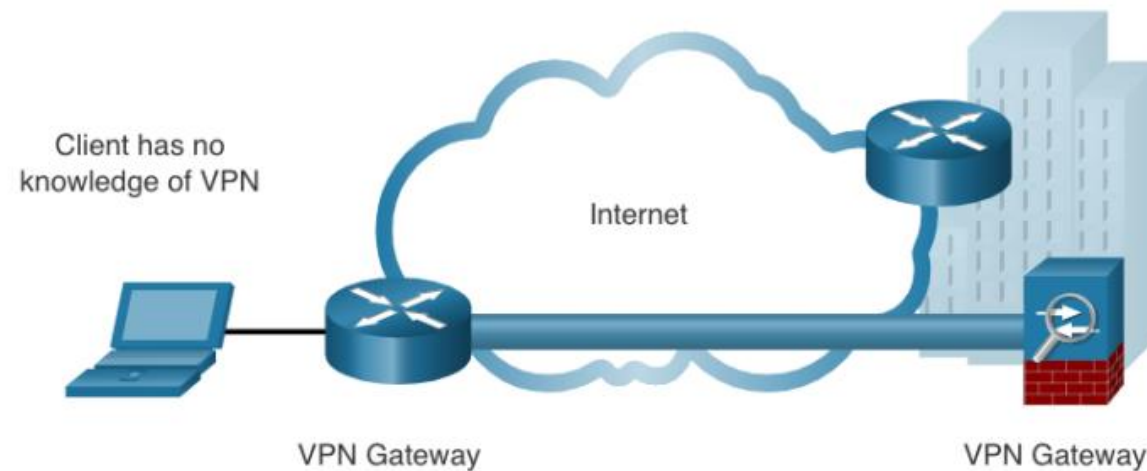
Client-based VPN connection. VPN client software must be installed on the remote user's end device.

VPN Types

Site-to-Site VPN

Site-to-Site VPN

A site-to-site VPN is terminated on VPN gateways. VPN traffic is only encrypted between the gateways. Internal hosts have no knowledge that a VPN is being used.



IPsec and SSL VPNs

Feature	IPsec	SSL
Applications supported	Extensive – All IP-based applications	Limited – Only web-based applications and file sharing
Authentication strength	Strong – Two-way authentication with shared keys or digital certificates	Moderate – one-way or two-way authentication
Encryption strength	Strong – Key lengths 56 – 256 bits	Moderate to strong - Key lengths 40 – 256 bits
Connection complexity	Medium – Requires VPN client installed on a host	Low – Requires web browser on a host
Connection option	Limited – Only specific devices with specific configurations can connect	Extensive – Any device with a web browser can connect

GRE over IPsec

Generic Routing Encapsulation (GRE) is a non-secure site-to-site VPN tunneling protocol.

- A GRE tunnel can encapsulate various network layer protocols as well as multicast and broadcast traffic. While Standard IPsec VPNs (non-GRE) can only create secure tunnels for unicast traffic.
 - GRE does not by default support encryption; and therefore, it does not provide a secure VPN tunnel.
 - A GRE packet can be encapsulated into an IPsec packet to forward it securely to the destination VPN gateway.
 - Encapsulating GRE into IPsec **allows multicast routing protocol updates** to be secured through a VPN
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GRE over IPsec

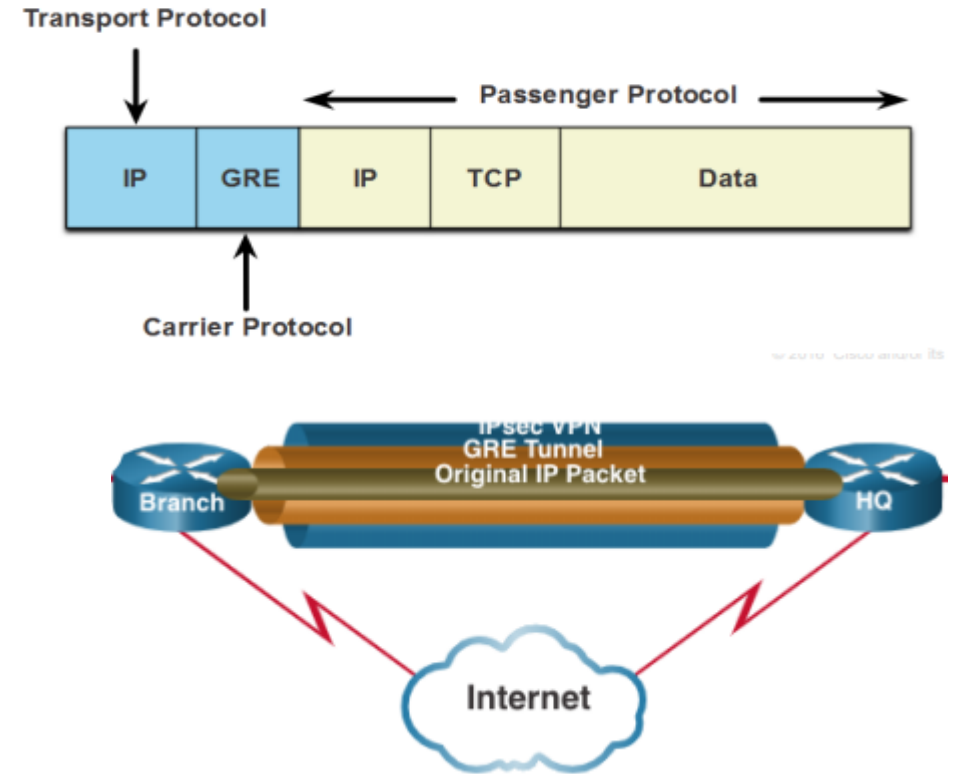
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Terms used to describe the encapsulation of GRE over IPsec tunnel

Passenger protocol. This is the original packet that is to be encapsulated by GRE. It could be an IPv4 or IPv6 packet, a routing update, and more.

Carrier protocol. GRE is the carrier protocol that encapsulates the original passenger packet.

Transport protocol. This is the protocol that will actually be used to forward the packet. This could be IPv4 or IPv6.



Dynamic Multipoint VPNs

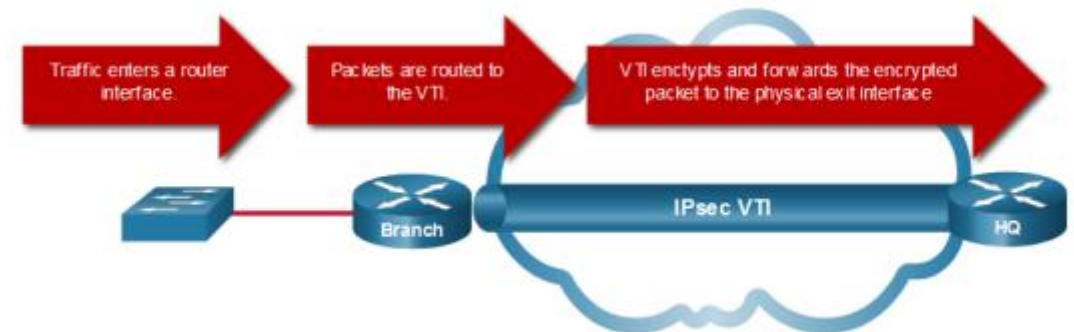
When the enterprise adds many sites, Ipsec VPN and GRE are not enough. Dynamic Multipoint VPN (DMVPN) provides a solution which is easy, dynamic and scalable.

- It uses a hub-and-spoke configuration to establish a full mesh topology.
 - Each site is configured using Multipoint Generic Routing Encapsulation (mGRE). The mGRE tunnel interface allows a single GRE interface to dynamically support multiple IPsec tunnels.
 - Spoke sites can also obtain information about each other, and alternatively build direct tunnels between themselves (spoke-to-spoke tunnels).
-

IPsec Virtual Tunnel Interface

IPsec Virtual Tunnel Interface (VTI) simplifies the configuration process required to support **multiple sites and remote access**.

- IPsec VTI configurations are applied to a virtual interface instead of static mapping the IPsec sessions to a physical interface.
- IPsec VTI is capable of sending and receiving both IP unicast and multicast encrypted traffic. Therefore, routing protocols are automatically supported without having to configure GRE tunnels.
- IPsec VTI can be configured between sites or in a hub-and-spoke topology.



Service Provider MPLS VPNs

Service Providers use Multi Protocol Label Switching (MPLS) in their core network. Traffic is forwarded through the MPLS backbone using labels. **Securing traffic between client sites is the responsibility of the service provider**

There are two types of MPLS VPN solutions supported by service providers:

- **Layer 3 MPLS VPN.** The service provider participates in customer routing by establishing a peering between the customer's routers and the provider's routers.
 - **Layer 2 MPLS VPN.** The service provider is not involved in the customer routing. Instead, the provider deploys a Virtual Private LAN Service (VPLS) to emulate an Ethernet multiaccess LAN segment over the MPLS network. No routing is involved. The customer's routers effectively belong to the same multiaccess network.
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IPsec

IPsec



IPsec Technologies

IPsec is an IETF standard that defines how a VPN can be secured across IP networks. IPsec is a suite of protocols

IPsec protects and authenticates IP packets between source and destination and provides essential security functions:

- Confidentiality
- Integrity
- Origin authentication and
- Diffie-Hellman

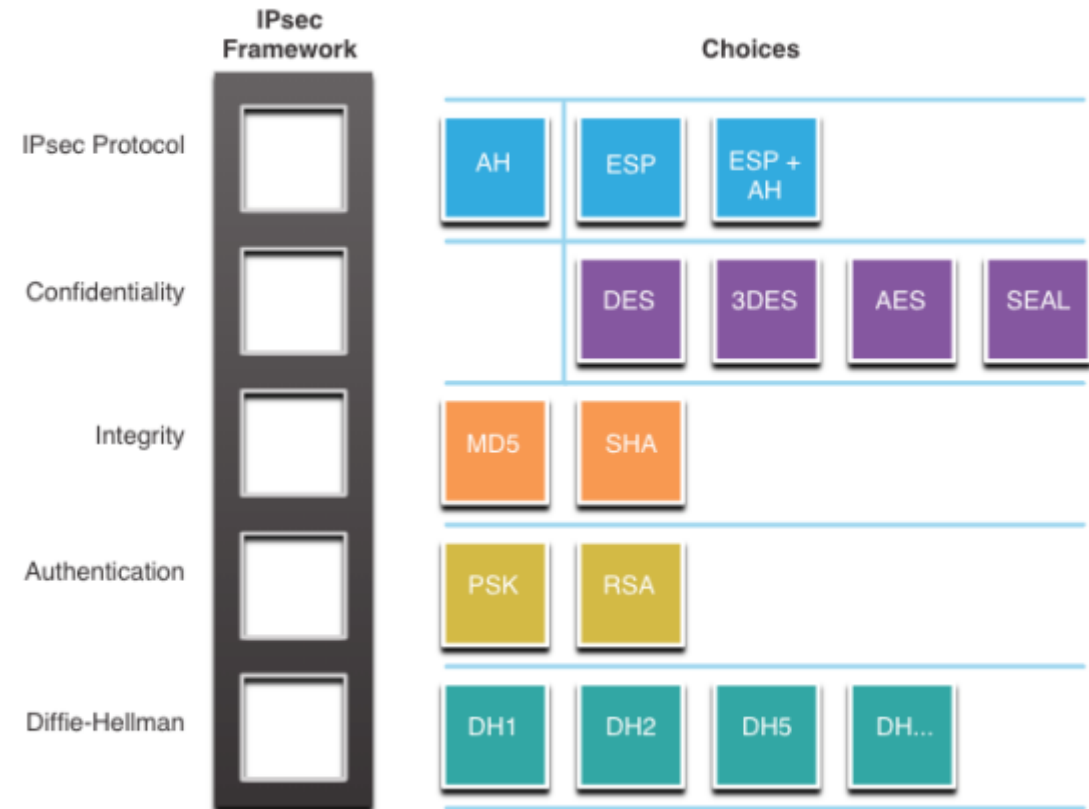
IPsec is not bound to any specific rules for secure communications.

IPsec Technologies

Cont'd

IPsec can easily integrate new security technologies without updating existing IPsec standards.

Open slots in the IPsec framework shown in the figure can be filled with any of the choices that are available for that IPsec function to create a unique security association (SA)



IPsec Encapsulation

Choosing the IPsec protocol encapsulation is the first building block of the framework.

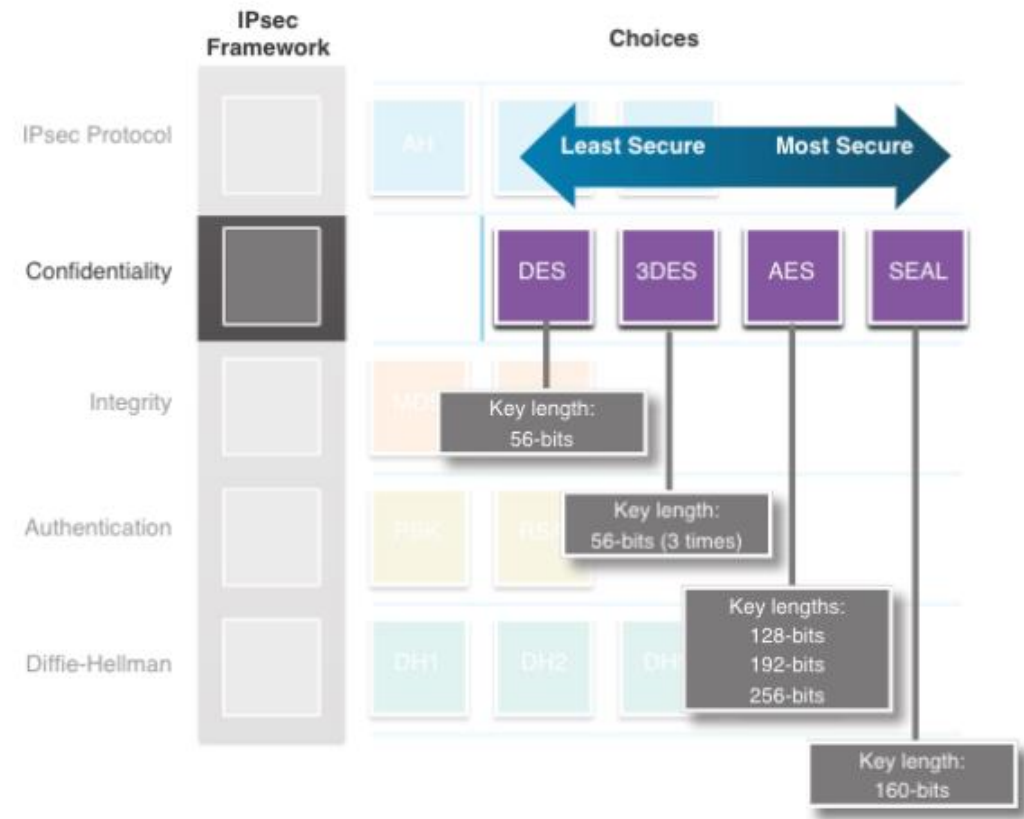
- IPsec encapsulates packets using Authentication Header (AH) or Encapsulation Security Protocol (ESP).
- The choice of AH or ESP establishes which other building blocks are available.
 - AH is appropriate only when confidentiality is not required or permitted.
 - ESP provides both confidentiality and authentication.



IPsec Encapsulation

Confidentiality

- The degree of confidentiality depends on the encryption algorithm and the length of the key used in the encryption algorithm.
- The shorter the key, the easier it is to break.
 - DES uses a 56-bit key.
 - 3DES uses three independent 56-bit encryption keys per 64-bit block.
 - AES offers three different key lengths: 128 bits, 192 bits, and 256 bits.
 - SEAL is a stream cipher, which means it encrypts data continuously rather than encrypting blocks of data. SEAL uses a 160-bit key.



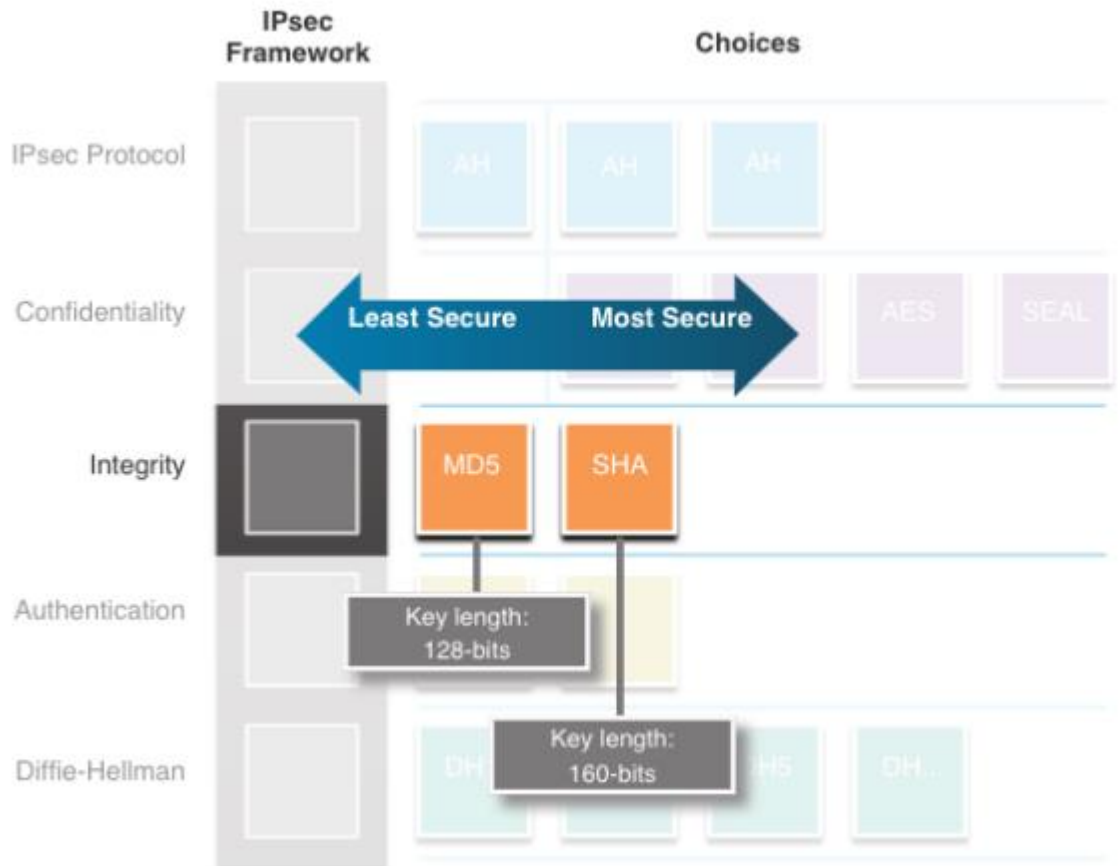
IPsec Encapsulation

Integrity

Data integrity means that the data has not changed in transit.

The Hashed Message Authentication Code (HMAC) guarantees the integrity of the message using a hash value.

- Message-Digest 5 (MD5) uses a 128-bit shared-secret key
- Secure Hash Algorithm (SHA) uses a 160-bit secret key.



IPsec Encapsulation

Authentication

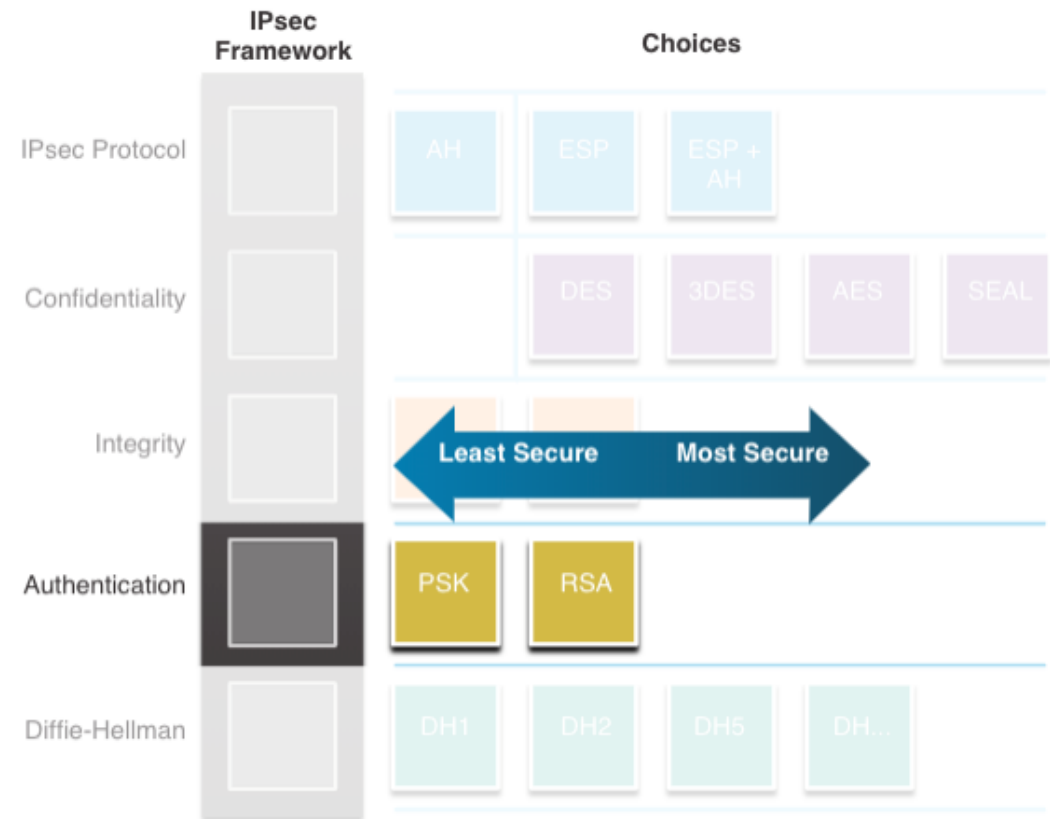
There are two IPsec peer authentication methods:

1. **Pre-shared key (PSK).** PSK value is entered into each peer manually.

- Easy to configure manually
- Does not scale well
- Must be configured on every peer

2. **Rivest, Shamir, and Adleman (RSA).** Digital certificates are used to authenticate the peers.

- Each peer must authenticate its opposite peer before the tunnel is considered secure.



Secure Key Exchange with Diffie - Hellman

Diffie-Hellman (DH) allows two devices to establish a shared secret over an unsecure network.

Variations of the DH key exchange are specified as DH groups:

Next Generation Encryption (NGE) is expected to meet the security and scalability requirements of the next two decades. It includes groups 20, 21 and 24.



Secure Key Exchange with Diffie - Hellman

Guidelines

- Groups 1, 2 and 5 do not provide an adequate security level against modern threats and should not be used to protect sensitive information.
- If you are using encryption or authentication algorithms with a 128-bit key, use Diffie-Hellman groups 5, 14, 19, 20 or 24.
- If you are using encryption or authentication algorithms with a 256-bit key or higher, use Diffie-Hellman group 21 or 24.

IPsec VPN Configuration

IPsec VPN Configuration

IKE Protocol

IPsec uses the **IKE** protocol to negotiate and establish secured site-to-site or remote access VPN tunnels.

IKE protocol is also called the Internet Security Association and Key Management Protocol (ISAKMP) (Only in Cisco)

IKE Phases

IKE Phases

ISAKMP separates negotiation into two phases:

- Phase 1: The two ISAKMP peers establish a secure and authenticated tunnel, which protects ISAKMP **negotiation** messages. This tunnel is known as the ISAKMP SA.
- Phase 2: It negotiates key materials and algorithms for the encryption (SAs) of the **data** to be transferred over the IPsec tunnel. This phase is called Quick Mode.

IKE Phases

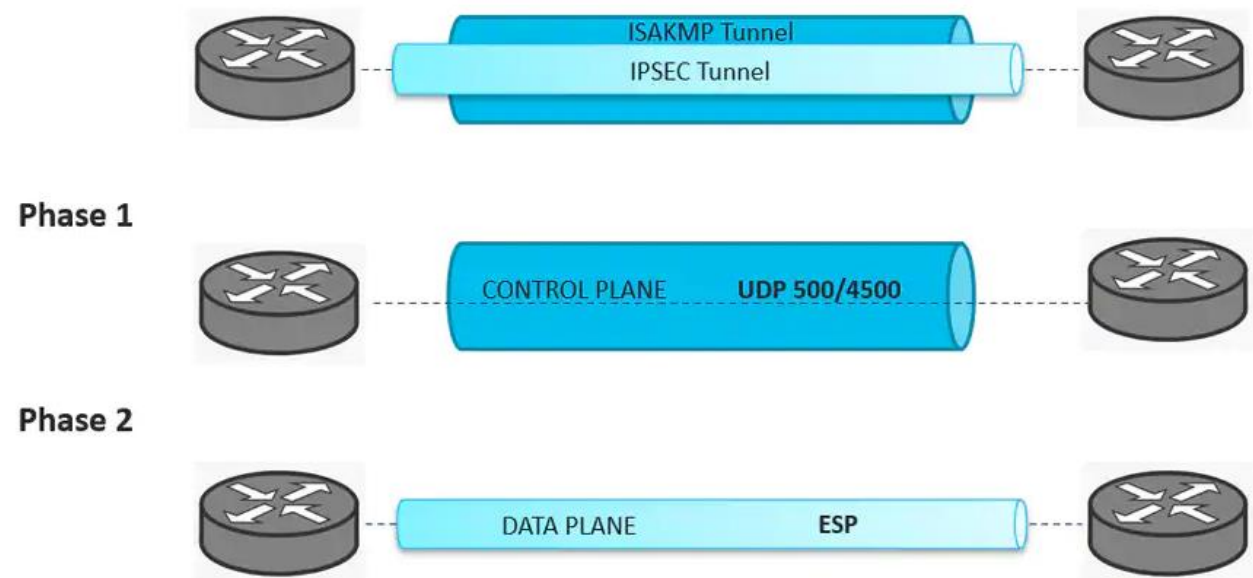
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Phase 1 tunnel is the Parent tunnel and phase 2 is a sub tunnel.

ISAKMP negotiation uses the UDP 500 and 4500 ports to establish a secure channel.

Phase 2 algorithms are independent of those specified in Phase 1.
The protocol used to encapsulate and encrypt these packets is the Encapsulation Security Payload (ESP).

ISAKMP-IPSEC Tunnel



IKE Phases

Cont'd

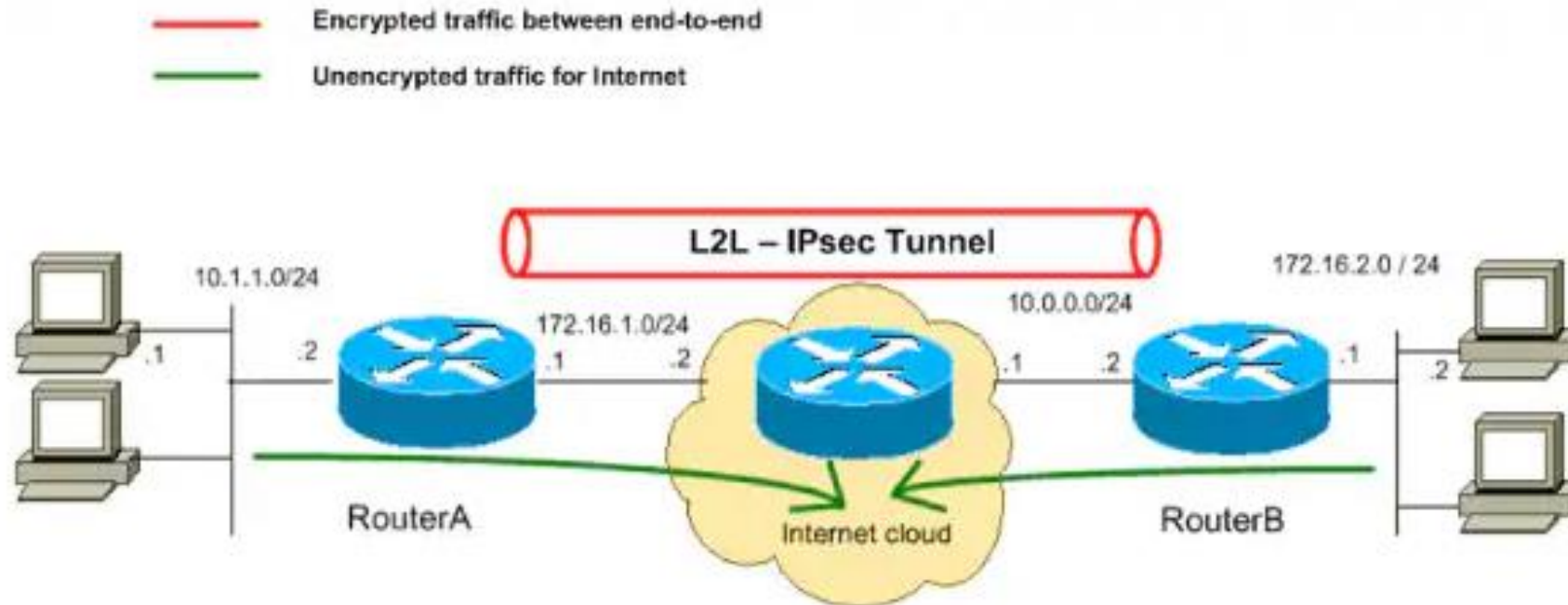
An IKE session begins when the initiator sends a proposal to the responder. The first exchange between nodes establishes the basic security policy.

The next exchange passes Diffie-Hellman public keys and other data. All further negotiation is encrypted within the IKE SA.

The third exchange authenticates the ISAKMP session. Once the IKE SA is established, IPSec negotiation (Quick Mode) begins.

IPSec negotiation, or Quick Mode negotiates the SA for the data encryption and manages the key exchange for that IPSec SA

VPN Reference Layout



IKE Phase 1 Configuration

Create an ISAKMP policy for Phase 1 negotiations for the tunnels:

crypto isakmp policy *priority*

Priority uniquely identifies the IKE policy and assigns a priority to the policy. Use an integer from 1 to 10,000, with 1 being the highest priority and 10,000 the lowest.

Under the Crypto isakmp mode configure the following options:

- **encryption (IKE policy)** ; default = 56-bit DES-CBC
- **hash (IKE policy)** ; default = SHA-1
- **authentication** ; default = RSA signatures
- **group (IKE policy)** ; default = 768-bit Diffie-Hellman

IKE Phase 1 Configuration

Configure a preshared authentication key using the **crypto isakmp key command** in global configuration mode.

crypto isakmp key *enc-type-digit* *keystring* **address** *peer-address* [*mask*]

<i>enc-type-digit</i>	Specifies whether the password to be used is encrypted or unencrypted. 0--Specifies that an unencrypted password follows. 6--Specifies that an encrypted password follows.
<i>keystring</i>	Specifies the preshared key. Use any combination of alphanumeric or special characters up to 128 bytes. Special characters include the following: !?"#\$%&'()*+,-./:;<=>@[\\]^_`~. (Type "CTRL-V" before the "?" symbol to avoid invoking help.) This preshared key must be identical at both peers.
address	Use this keyword if the remote peer ISAKMP identity was set with its IP or IPv6 address.
<i>peer-address</i>	Specifies the IP address of the remote peer.
<i>mask</i>	(Optional) Specifies the subnet address of the remote peer.

IKE Phase 1 Configuration

Create an ISAKMP policy for Phase 1 negotiations for the tunnels:

```
crypto isakmp policy 10  
  encryption aes  
  hash sha256  
  authentication pre-share  
  group 14
```

Specify the pre-shared key and the remote peer address

```
crypto isakmp key vpnuser address 10.0.0.2
```

IKE Phase 2 Configuration

Transform Sets are used to define IPSec security associations (SAs). IPSec SAs specify the IPSec protocols to use to protect packets.

Use the **crypto ipsec transform-set** command in global configuration mode.

crypto ipsec transform-set *transform-set-name transform1 [transform2] [transform3] [transform4]*

Up to four “transforms” can be used: one Authentication Header (AH), one Encapsulating Security Payload (ESP) encryption, one ESP authentication, and one compression. These transforms define the IP Security (IPSec) security protocols and algorithms.

Accepted transform values are described in link below.

<https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/security/a1/sec-a1-cr-book/sec-cr-c3.html?dtid=osscdc000283#wp2590984165>

IKE Phase 2 Configuration

Create the Phase 2 policy for IPsec negotiation.

```
crypto ipsec transform-set myset esp-aes esp-sha256-hmac
```

Create an ACL for the traffic to be encrypted. The traffic which does not match the access list is unencrypted for the Internet.

```
access-list 100 permit ip 10.1.1.0 0.0.0.255 172.16.2.0 0.0.0.255
```

The Actual Crypto Map

Create the actual crypto map.

crypto map [*ipv6*] *map-name seq-num ipsec-isakmp*

ipv6	(Optional) Specifies an IPv6 crypto map. For IPv4 crypto maps, use the command without this keyword.
<i>map-name</i>	Identifies the crypto map set.
<i>seq-num</i>	Sequence number you assign to the crypto map entry.
ipsec-isakmp	(Optional) Indicates that IKE will be used to establish the IPsec for protecting the traffic specified by this crypto map entry.

The Actual Crypto Map

IPsec crypto maps define the following:

- What traffic should be protected
- To which IPsec peers the protected traffic can be forwarded--these are the peers with which an SA can be established
- Which transform sets are acceptable for use with the protected traffic
- How keys and SAs should be used or managed (or what the keys are, if IKE is not used)

The Actual Crypto Map

Create the actual crypto map.

Specify an access control list (ACL), which defines the proxy identities (local and remote host/networks).

```
crypto map mymap 10 ipsec-isakmp
set peer 10.0.0.2
set transform-set myset
match address 100
```

Apply the crypto map on the outside interface.

```
interface GigabitEthernet0/0
ip address 172.16.1.1 255.255.255.0
crypto map mymap
```

Verify The VPN

Some commands to Verify that the VPN is working.

show crypto ipsec sa. Shows the settings, number of encaps and decaps, local and remote proxy identities, and Security Parameter Indexes (SPIs) (inbound and outbound) used by current Security Associations (SAs)

show crypto isakmp sa. Shows all current IKE SAs and the status.

show crypto session remote <IP address of peer VPN endpoint> detail

Verify The VPN

Some commands to Verify that the VPN is working.

show crypto map. Shows the crypto map structure created with:

- Name of the crypto map and sequence number
- Peer address
- Name of the ACL applied along with the local and remote proxy identities
- Values of the IPsec transform-set used
- Interface on which the crypto map is binded

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Thank You

Thank You

Appendix

Additional Concepts

Secure Key Exchange with Diffie - Hellman

Diffie-Hellman (DH) Groups

- Diffie-Hellman group 1 - 768 bit modulus - AVOID
- Diffie-Hellman group 2 - 1024 bit modulus - AVOID
- Diffie-Hellman group 5 - 1536 bit modulus - AVOID
- Diffie-Hellman group 14 - 2048 bit modulus – MINIMUM ACCEPTABLE
- Diffie-Hellman group 19 - 256 bit elliptic curve – ACCEPTABLE
- Diffie-Hellman group 20 - 384 bit elliptic curve – Next Generation Encryption
- Diffie-Hellman group 21 - 521 bit elliptic curve – Next Generation Encryption
- Diffie-Hellman group 24 - modular exponentiation group with a 2048-bit modulus and 256-bit prime order subgroup – Next Generation Encryption

Next Generation Encryption

NGE of cryptographic algorithms include:

- Symmetric encryption -128-bit or 256-bit Advanced Encryption Standard (AES) in GCM (Galois/Counter mode)
- Hash - Secure Hash Algorithms (SHA)-2 (SHA-256, SHA-384, and SHA-512)
- Digital signatures -Elliptic Curve Digital Signature Algorithm (ECDSA)
- Key agreement - Elliptic Curve Diffie-Hellman (ECDH)

Elliptic Curve Cryptography (ECC)

Elliptic Curve Cryptography (ECC) is a PKC (Public Key Cryptography) algorithm based on the use of elliptic curves in cryptography. It is designed for devices with limited computing power or memory to encrypt internet traffic. A common use of ECC is in embedded computers, smartphones, and cryptocurrency networks like bitcoin, which consumes around 10% of the storage space and bandwidth that RSA requires.